ActInf Textbook Group ~ Cohort 7 ~ Session 11 (Chapter 4) ~ 9/23/2024

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okay welcome back cohort 7 we're discussing chapter 4 so anyone can bring up any comment or anything they remembered or a question from four or some other comment from earlier on uh or we'll look at the chapter or look at questions but first does anyone have a comment or just any other anything else they want to kick it off with okay type in the chat or or raise your hand or anything um chapter 4 has two roads leading into it with chapter 2 and three so this is really where we see active inference generative models after the more General context of one and then the low and the high road in in chapters 2 and three um anyone like is there a part of four that anyone wants to go to or just a quote that's interesting just to begin okay 4.1 single paragraph describes in very concise kind of llm like style what the next sections are going to be uh section 4.2 goes from base equation base theorem exact base which is exactly how you update the prior into the posterior given an observation that's a calculation that is done like in one Fell Swoop the idea of using some bound on exact basing inference so the idea of doing approximate basing inference using some other proxy not surprise exactly but abound on surprise like variational free energy the idea here is that that bound could be picked to have certain properties like be able to optimize it incrementally with gradient descent like in machine learning like the title of the section they do start with base theorem though this chapter also says that it will present the active inference generative models and it does there are a few kind of background detours or or um interludes like Jensen's inequality towards equation 4.2 uh they use the definition of prize as log likelihood and then through the usage of that log or just any function that has like a decreasing uh rate of decrease of the slope but never turning back negative um the log surprisal allows like a um strict bound to be described which is called the free energy or negative free energy like just depending on the negative sign um okay Jonathan a previous uh participant and a math teacher I think Professor has written this document let's just see it for a second some of these it's like okay I think this date is automatically chosen but that would be interesting if he updated it so recently let's look through this while there are some superb explanations in the book there are important details left out in the derivations which may not be that important on the first pass but I think make things much Clear when you want to understand it in detail our general end will be to observe things we expect to observe we will come on later to the things that we want to observe erve when we think about preferences but for now we can just think about the things that a model predicts that we are likely to see here's the part with Jensen's inequality so exact base 2.1 just like in the book Jensen's inequality this is going into more lines of detail from what is in the book this gives some um lines to follow this bottom line here here's surprisal log likelihood this is the likelihood of that observation and then log likelihood for for a variety of reasons to to get some 01 uh interval properties to get that boundedness um to be able to compare multiple uh like uh orders of magnitude of probability so this is whatever it is here's the K Divergence between like the distribution that you can control your surprisal from and uh the empirical distribution P um chapter 2.5 or chapter 2 equation 2.5 had the variational free energy and there was multiple lines it seems like he's going into more detail here connecting how free energy described in Chapter 2 and and then expected free energy in equation 2.6 and then connecting it more to chapter 4 so that's that's also kind of a background math like a standalone math thing any ideas or questions on 4.1 or 4.2 so 4.1 again single paragraph yeah Jeff go for it yeah no um I was recently reading a paper and actually talking to Hector Dean and Magdalena um about um how the decision making process you know binary tree branching it doesn't scale it needs to be bounded right like we can only make one decision at one point in time but we're Li limited by our you know by the body that we inhabit and our signal processing when you H when you have systems of systems who are you know a scale-free right uh nested agents right who were interdependent they they loss and compression there in order to isolate the signal from the noise and also you know have a pruning function for that that that that [Music] um decision the the decision tree that's just where my mind is right now thanks Jeff yeah these are good um you you point to these kind of constraints on the inbound with the signal processing and then the the constraints on the outbound I mean the ultimate constraint on the outbound one embodied body being able to only do one like posture at once so it's this tension between okay you could plan out all these multi-joint actions couple of joints a couple of affordances a couple of time steps that could be like all the atoms of the universe and then so there it's it's it's really interesting because Seth Lloyd in programming the universe which he wrote back in 2006 he shows very clearly that like you know a a game board not big not much bigger than a than a go set has more potential States than all the all the matter and energy in the universe right so like you know in one run of the universe you will not explore all the potential states that could possibly exist yeah and I I I don't know about the branching uh like Multiverse but sometimes it's like that's just like saying it's kind of acceptance of that whether whether as a uh whether that's an epistemically strong stance or whether that's a a cope but it's just like saying yeah that much branching really does happen it's like okay but it doesn't really get around that like chessboard explosion and then I mean that just is essentially an empirical validation that cognitive heris discs are used for sense making and decision making yeah even like when you model like very small sequential domains um it becomes very unwieldly very quick as you scale the board one one actin uh connection there is this uh branching time active inference Direction and maybe a little bit of more recent work this is this is uh drawing methods from uh dealing with these tree and combinatoric explosions whether that's whether you think about that in the combinator explosions of interpretation on the inbound or like action s joint action planning uh or time or plan that kind of tree search has been a classic problem like chess algorithms and and deep blue and Beyond so yeah and alphago and Alpha star and whatnot right they' they've solved these for a specific domain um exactly and and broadly some of those approaches are Monte Carlo based Monte Carlo is like a city where gambling happens so these are the gambling sampling based approaches and then an alternative complimentary approach like some problems with it both neither one or the other do well Etc but another approach is to cast that as a variational inference or as a basing inference um problem and then use very ational methods to approximate the posterior whereas Monte Carlo samples and and Lance daosta has done a lot of work on the sampling and the variational uh approaches but that that's yeah a few other random thoughts like it's it's easy to make a a cognitive behavioral model that's like it's just like maps that don't to territory as well for for one reason another mean you could make it wildly overpowered like you could have it be a a mouse foraging model with a gigabyte of short-term memory of the GPS location it's like the model won't complain that mice don't actually remember the GPS coordinates so it um H taking into account those constraints is often like the hard or a hard part of empirical modeling whereas if one is just more interested in like a looser like bio inspired approach and then is seeking performance or some other attribute just Loosely based upon biology then of course there there's there's nothing holding back but to actually connect and make useful predictions for a given like biological system then how how um that gets represented becomes challenging because it isn't just about making like the highest performance model because it's more about fitting to empirical data there's many ways in practice to to deal with this ranging from core screening data to picking simpler variational distributions using uh different kinds of model architectures so that there there aren't like wide computations like um the all by all by all like you just figure out other ways to structure the problem so that the um computational proper ities are not so disadvantageous if if it really even is a problem at that scale um there can be major heris STS like um if the input data were 4K video I mean one approach you could take that in as a huge Matrix with like millions of numbers and then in that case maybe it is hard to run on but then maybe um another approach like uh passes the video frame to an llm and then gets back estimate on which items are in it and then takes in that list of items so there's a lot of ways to to deal with the and and and also just to study okay how does the runtime change as a function of like the size of the input or the number of sensors like is it linear as adding more sensors or sublinear or is it super linear 4.2 is is a another background but a really important intuition figure this sets us up to look at the base graphs that will be like the PDP and represent the perception and action components but um happily this is is not unlike a uh systems dynamic or systems modeling or just kind of like mind map or flowchart representation I mean um if you just talking about day-to-day situations like there was a lightning strike and then there was light and sound I mean maybe somebody wouldn't draw the two and the three boxes just draw X directly connected to Y and to W but topologically this is pretty much what is is conversationally described when people are talking about um observed and unobserved phenomena any random thoughts on that or or like just things people seeing this or about these base graphs overall I saw the muszle muzzle flash and then I saw the impact similar yeah here that's like this one yeah that same idea that okay nodes are going to represent distributions being between zero and one like probability distributions um or just random variables like it could be number between one and 10 or could be um just different kind different kinds of uh support for those variables different values that they could take on uh so it's kind of just like a variable in a program um that same idea that variables are declared and then their their connections and influences are represented with edges and some of these are observables the O's meaning that they're kind of Taken even if they're understood to be a noisy sensor still the observation is taken as it is um and then unobserved V variables like the latent States and then the the sort of connection between signal processing on bottom and control theory on the top and then that work is done with the B Matrix three which is the probability function that maps The Hidden State like the real temperature in the room at the next time Point conditioned upon the current time point and the current behavior so this is a uh action conditioned Markov transition Matrix it s is like a vector it could be like one temperature data point in the room or it could be like a vector of of thermometers this B is like a stack of index cards each card is s by S Dimensions it just describes from and two so if you don't have any affordances or if there's just one affordance like this the the just this downstairs part of the system is unfolding with with no action selection then it's just like from and two and that's like a Markov trans position Matrix and then policy selection is just selecting which of those index cards is going to be used for picking uh which the transition to the next time point and then kind of a a simpler but analogous role is played by the a matrix which can be used to go from a given hidden state to a distribution over outcomes so it's like I know that my thermometer has plus or minus one you know uh measurement noise so if it were really 81 then I would expect this distribution between 80 and 82 that's using it in the forward Direction like generating synthetic data and then also there's taking in a point measurement and then updating your prior to what your best belief is now H not shown here but that's basically where attention comes into play if you are not attending to the observation then it has no impact if you have complete attention to the observation then it overrules your prior and you update your belief to the observation so that's like the Precision or the kind of um sharpness on a and and and how it relates to the strength of the product figure 4.3 uh shows on the top the discrete time setting and on the bottom The Continuous time setting the the kind of uh structural similarity is highlighted like with using the same box numbering for the action selection State transition and the sense making component one one three and two boxes at the same time different notation letters are used like o for the observations indry time and Y um for continuous time highlighting structural similarity with generative models whether they used screet time or continuous time while also highlighting their differences and the major difference is that in discrete time discreet specific time steps are considered like s of t t plus1 t plus2 for a given Horizon whereas in the continuous time setting Which is less about like transitions between states discrete States and more about like a continuous differentiable line passing through the present this is more analogous to the Taylor series expansion so there are not actually future time points specifically predicted the only specific uh evaluation of the function is at the the current time point and then the past and the future get fit with a tailor series being able to fit further and further out with more and more approximations so plus here for for for uh one low cost you if you do have a prediction for every past and future data point the downside is it's hard to even know how relevant that prediction is so like here we take the first uh we take the first pass which is just evaluating it at the current moment that's just this point then you take the first derivative that point and you get the red line so it's like okay so it's going up in the in the right and it's going down the left but then let's say you expand your approximation and now you're in the third uh powering but e even though in the area around zero it still has the same slope but now there's a totally different story with it going off to Infinity quickly to the left and then down to the right and then go into the fourth powering and and it's maybe it's a different story again like with coming so even though again it's all being guaranteed to to get more and more accurate in the neighborhood but but uh knowing even getting a Precision on well how close is my prediction like to the real one that can be challenging to know so then it is an empirical question like how good of approximation is this Taylor series expansion this gets Revisited in chapter seven and eight so I mean this is kind of the main generative model description of the book since two and three get us to the generative models and six is about how to design them and five is about some specific ones that are used and then nine about getting connected with the data this figure and then chapter seven and 8 which it sort of anticipates are the main generative model descriptions so this is like a micro 7 yeah go for col yeah so that that applies also to the action of doing nothing right so on the different potential policies one is always doing nothing right so that if I do nothing for instance it doesn't matter if it's in the continuous or or the discrete approximation um um you you you can refine your inference either by making continuous observations by doing nothing even if you don't interact with the system right or or the system itself could could without doing anything from your side could evolve and they naturally evolve so that applies also to that policy to the policy of doing nothing right is correct yeah great great comment like um H how is doing nothing dealt with so sometimes let's let's just say we're in a uh grid World movement sometimes do nothing is is incorporated as an afford so there's like up down left right stay another approach would be um higher higher level stay or or move and then lower level up down left right so those either those could be used um and I mean you you point to a really important question which is the control theory and Game Theory it's often just framed like it's obvious when action should be taken and so in a turn-taking game like chess you you could talk about like well you know when should this Pawn you know advance or something like that but that's that's within the game time um it's a it's an alternating turn taking but then when there's a system that's changing external then like I mean like a an economic Market it's like timing the market there there might be way broader changes that your actions are less influential on such that for for um for achieving like pragmatic or epistemic value it it almost matters less which action is done but more about like when done so those that's like kind of another flavor of of control theoretic question but it's not really so easy to address in the turn-taking game setting but it is more kind of amenable to to ecosystems where things are really happening yeah more Dynamic way uh ones in which you know you you cannot infer the state just by one observations you need to wait for a number ofs they come along in order to be able to make a good inference for of of the current state yeah I mean that's that's even another angle which is like let's just say that um measurements cost us one like there's just a cost per measurement um so we have a budget so we can't just you know get a million measurements in in a split second but then also there's like a rate of change of the underlying system this is like buying stock market trading data or like planning and experimental data so then there it's this assessment of like well for deciding on what to do that's like the actionable side what is the tradeoff of like which measurements and when to make those measurements yeah and yeah some of these are are would be really interesting to kind of to kind of sketch out and just show how different heris sixs would lead to like different kind of um biases in the sense making and decision making like if you just if you in if you if you have a a c you have a certain number of credits for how much measurements you can make well if you immed immediately pursue the highest Precision then you're initially even if it is initially accurate then that assessment would become inaccurate if you wait until the very end you cross the finish line like with the highest accuracy but then for for most of the time you had the least of an idea well if if you space it out then like you may have a lower level but but above nothing so then like that kind of comes back to this constraints topic and and how the interesting pieces about how do you how do you model those constraints because that's where the the decision like the kind of wide open space that's why it's so hard to predict what an eoli I mean or a mouse will do yep yeah but but yeah definitely about doing nothing is as that's that like or or I'm just thinking of in in like an agent based model sometimes there even is no Stay option so then there's like a lot of movement just moving between two squares back and forth like if they're already on kind of a local Maxima or something but it's like but that that kind of behavior isn't really seen but that just because there isn't the Stay option but just like walking around like the bird is just like sitting on the tree not like spending not resting by hopping back and forth but that's like a that's a that's a huge difference but that's just about having that's just about the structure of the model and how it's being fit yeah than yeah also it highlights how like free energy is very model specific so just whether thinking of the perceptual variational free energy calculation like just the sense making from noisy and partial data or the action selection as an inference uh problem that free energy calculation is based upon the data the family of variational distributions the generative model so then that it it yes that's an instrumentalist perspective but it it highlights why like well what's the free energy of that squirrel why is it sitting on the ground like that it's like you're quite a ways off but if you described okay the data or the squirrel's location the variational distributions of the gine that I'm making and the prior and the likelihood on location for the squirrel here's why it is or isn't surprising to me why my model is returning a high or a low free energy for the squirrel being there but that's more of just describing your statistical model but but just a a a sort of qualitative description of free energy as if it was just naturally an unambiguous part of a of a biological systems kind of a stretch I'm reminded of the halting problem like how like you can you know write endless loops kind of kind of reminds me of uh you know hoffstead or two you know good aler and Bach and uh uh I am a strange Loop while at uh I'm gonna welcome back to the this was uh uh in this this Workshop over the previous days Chris Fields gave us talk I'll put it in our in our uh page and I'll put it in the chat and it just sort of there was it was just some interesting discussions like well you know never step in the same river twice we're we're never the same we're we're we're whether you take that from a material like breathing CO2 and oxygen um food um or just information and like planning and stuff like that it's like we're never the same so then what this part was really interesting the neural like the I don't know if this comes from the icade or from the motor like analogy function or something like that but the embodied math and and just the the way that the category the and I think other equations get um interpreted then it's like so then what is the same it's like well the identity is what is the sameness but then it's like but then the but then your identity When You're 15 when you're 30 it's that is different so it's just like I I I I I don't think his his DOA answers it per se but it gives some some interesting stick drawings to begin exploring like that sameness and differentness which is one of the most kind of pervasive crossmodal aspects of experience yet at the same time it's it's it's really not so simple when the our identity materially and cognitively and metacognitively is changing but it does have a sameness too and he mentions octm and uh connects a little bit to the some he doesn't imagine it but but the the the um physics is information processing but just showing like the sort of uh um usage of a of a very ational free energy or something like that for like a phenomenological human State this is just prompted by the the geser bach it's like the whole point is that's a strange system it can't just be taken for granted like even what direction what layer of what it that's why there isn't like an open source kernel for for the human mind it just it it it it seems to be so baffling and strange again revisiting equation 2.5 and 2.6 variational free energy F of Q and Y it's with a belief distribution and a new data point so it's just like a real time um real time through time minimization it can include it can be a sense making only or it can include like instantaneous action inference so without explicit Planet single time step updating of beliefs including beliefs about action whereas if you want to get into explicit planning than expected free energy is used this is a lot like equation 2.6 still in the discreet time section this is just going through more details of stuff that's going to come back in chapter 7 4.5 continuous box a blanketed off box on marov blanket um with its definition Upstream causes of X Downstream effects of X and the co-parents of the children of X after that short definition immediately going into these two methods kind of like Network propagation or percolation methods for having messages passed among nodes that implement this variational uh gradient descent here the uh it's shown with kind of another notation set um the left is showing a a sort of classical predictive processing hierarchical model so observations through time are coming in O Tilda each of these four that look closer together are one layer of the hierarchical predictive processing like hierarchy um at each level there's um a descending prediction that's getting contrasted with the with the ascending observation and then that is getting like turned into an error residual it's getting passed back up to a higher level um so here's like primary sensory processing level um another nested level and then I think just the error unit for like an overall so this is like measurement coming in this is like the ones place this is like the 10's place and then this is just like green light yellow light or red light how accurate is the 10's Place prediction that's so that's a snapshot or or kind of uh Through Time emphasizing the hierarchical predictive architecture whether or how the brain does it is secondary to here it's just this is a modeling Motif that's used in the basian systems here on the right um is looking more through time so here just like in the top of figure 4.3 it's observations coming at the current time point and the past and the present uh past in the future time point and then just showing a little bit of like kind of the information flow of how those updates get pasted um in order to update beliefs about policy including like and updating hidden States conditioned on policies it it it says it emits some intermediate connection so it's like not exactly sure um how much to look into the edge connections but that the it is fully described but here they they don't show all the edges here now we're kind of back into this this continuous time case this is a really important set of equations the SPM statistical parametric mapping or like anything involving neuroimaging this is kind of a classic pair as well as in physics so neuroimaging this is the observations coming in with like a signal and a noise could be a sensor noise and then x dot is like a spatialized flow landscape of an underlying State like neural activity so the the measurement of fmri might be like the um just the intensity of the signal coming from a given voxal at a time that's the observation which which is due to it to underlying like blood flow changes and also noise and then there's this underlying neural activity landscape which is x dot so here's a situation where they're both normal uh distributions 4.2 generalized coordinates of motion is the tailor series you can take the tailor series or generalized coordin motions on the observation and on the latent State x dot so the these are derivatives first second Etc derivatives and and and again in the kind of classic hierarchical predictive processing scheme that the the Precision comes into play llao approximation free energy may be approximated by a quadratic expansion around the posterior mode so the mode is the most common like the highest point on the probability distribution the median is the 50th percentile data point the mean is like average value the mode is just picking the highest point which is like the single maximum likelihood estimate and then the llao approximation is taking um to the second power around that point so you have a quadratic curvature and then fitting that quad ratic distribution so if if if it's not a if there's not a central tendency then of course this will still only be able to fit a quadratic so if it's like three humps then you're either going to have a really over dispersed quadratic or have one that just fits over one but um it's an approximation now it's again the hierarchical and the through time but instead of the O's it's the Y's so now it's kind of swapped into the notation set of the continuous time that's chapter 4 five is gonna gonna do some reviews on neural systems which is where these models have been applied most but yeah four is where the the generative models at least the simple motifs for discret and continuous time are shown and there there's there's other motifs and patterns like hierarchical and things like that but um they highlight the difference and similarities between the continuous and discreet time and that's like a big structuring element of the book with chapter 7 and 8 Daniel just a question uh I haven't seen yet any uh real world problem that has been using this type of hierarchical models everything I've seen so far is flat can you give me some examples of references links to uh works that use hierarchical model please yeah good good question for for this robot navigation paper describes um okay it's not an Open Access paper here's their hierarchical model so this is one setting in a multiscale movement like locations in a warehouse and then robot body postures that's that's like one where the higher and the lower level are both um like mechanical and positional um the hierarchical hybrid model with continuous on the lower level and discret on the top is kind of the folk psychologic iCal template that's that's that um in this Smith at all paper I mean they argue this is basically the um implicit or explicit generative model and people talk about in a folk psychological sense let's see if they have any I don't know if they have a visualization but um and then I think um but okay that those are a few um in the implementations I agree they're usually higher they're usually single um layer uh it could be a range of things ranging from Advanced models um happening uh proprietarily another option is that a model like um for example Alex Aria's work and uh these kind of the neural uh generative coding this this does make hierarchical models so there's a flatness because it's like a neural topology a neuromorphic system but then the calculation it implements could still be hierarchical and I think another um interesting direction to explore djar hner dream to control Information Gain in the planning objective Carl first mentioned sometimes dreamer is like um some variation of active inference so I I think the textbook clear clearly shows the model doesn't have to be hierarchical like architecturally or functionally for it to be active inference even for for perhaps interesting or multiscale behavior to arise definitely showing more clear nested models though will be interesting and understanding like okay if you have to do 100 time steps let's see the 100 times step planner the 10 times step planner the decade and then the the ones place planner the no planner the single times like kind of create in all of those alternative ways to to parse the problem I think would be very informative because there's probably niches and inputs and and uh sense making and action problems for which like but there's no free lunch but but on on some data set each of those you could imagine could be effective or not but for real world settings it's definitely an empirical question like how because if if it's just about like well the world the systems are hierarchical or they're nested but then that's just getting into the the map being the territory I'm has anybody ever framed it in such a way of the so so like reinforcement learning and deep learning right we're getting these massive data sets and we're trying to figure out how to make an agent move about the world and it feels to me like the framing of act active inference as a way for an agent to move the world around it almost like comes to mind so instead of moving moving around a bounded environment right the the the agent is like moving the whole like the agent is the Central folky of existence I I don't know what's been done but sometimes that's called like egocentric navigation where the ref point is whereas like in the grid world examples that we look at in pmdp the agent like Square two comma two but it's like but how' you know you were in a 3X3 maze whereas for an open space than than the egocentric navigation and then it also connects like to this um to the the action and non-action earlier because it's like um I mean one nistic of movement and and plausibly something that is done for movement or like being in the car it's like well you could think about the that the world and and its causes are fixed and then you're moving over that or it's like being on a treadmill it's like yeah the freeway is like passing forward so then from the egocentric frame that's why it feels stationary even like being on a train but then from from an allocentric frame you might think about yourself moving positionally across a fixed space so and again that goes to like making these animal behavioral models that are just like wildly off base like people will for example set up an L-shaped wall and then they'll test okay well do do animals have an internal space representation like will they take the hypotenuse cuz it's shorter to get to the food or or will they just follow the edge but it's like but the animal may not even have the vision to see across the diagonal or they may just have a a habit to stay up against a wall because it's safer so it's like you but so then having made a made a um allocentric model where it's like okay it's going to calculate a hypotenuse and then it's going to minimize the path distance it's like it's just has nothing to do with the her istics of the animal so then then then what whatever the P value you get there is it's just like it's a model output but it it doesn't help address like an interesting question about the animal in in such a clean way you know I'd be it'd be really interesting to explore this in terms of the Interior changes that occur um in the predictive models of of animals that undergo metamorphosis like um you know a caterpillar to a butterfly like or um the uh spawn of um Barnacles right like they they start as a U mobile little and they can swim about the ocean and then they fix onto something and then like they're fixed and the world then is anchored for them yeah also like the I mean the multi-generation like monarch butterfly are they using a are they just you could imagine all of the you could put a GPS tracker Etc think of them moving through space or it could be a pure heris based upon like light and magnetism and seasonality all these things that doesn't require planning but then back again back to this combinatorics of planning it's like well if it's combinatorics of every muscle twitch it's like well if if that's how if if yeah that's maybe not what is happening but this is this is the other map territory what and the Butterfly is able to do that but but can't NE necessarily emulate chess so even just getting the trick of detering complete computer to get constraints and capacities that even resemble the biological system is totally not a given and the and the model won't throw up a flag again if you give it a you give the butterfly a GPS coordinate and and and a Bluetooth communication with the rest it's like you can make that model but it doesn't have the biological plausibility okay thank you see you next time thanks bye byebye